

Lecture 2

Decision Theory

Graphical Models — Inference Stage

Decision Theory

问题: Once we have the joint distribution, e.g. $p(x, c)$
how we make a decision?

Example: Cancer screening from chest X-ray

input x

output c

$\begin{cases} C_1 & \text{cancer present} \\ C_2 & \text{cancer absent} \end{cases}$

Misclassification Rate

问题: 给一个 new x_{new} , 我们怎么判断他的 C_k ?
 $k=1, 2$

目标: 我们要先确定我们想要最小的 objective function, 才能优化

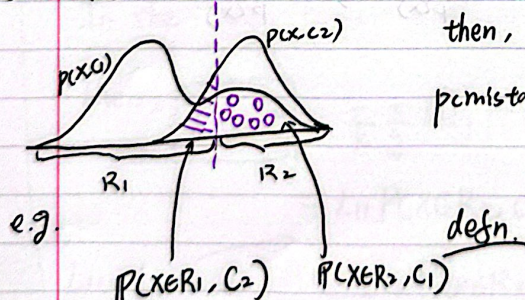
In this case, we want to $\min E[\frac{\# \text{mistakes}}{\text{total}}] \leftrightarrow (E[\frac{1}{n} \sum \text{mistakes?}])$

$\Leftrightarrow \min P(\text{mistake})$

$\Leftrightarrow \min [\text{Misclassification Rate}]$

最终我们发现: $\min P(\text{mistake}) \equiv \text{选概率最大的}$

Pf: Divide input space into region R_k s.t. all points in R_k are assigned to class C_k .



then, in this case

$$p(\text{mistake}) = \int_{R_1} p(x, c_2) dx + \int_{R_2} p(x, c_1) dx$$

$$\updownarrow \\ P(x \in R_1, c_2)$$

$$\updownarrow \\ P(x \in R_2, c_1)$$

then, Misclassification Error:

$$P(\text{mistake}) = \int_{R_1} p(x, c_2) dx + \int_{R_2} p(x, c_1) dx$$

Finally, we have

$$\min_{R_1, R_2} P(\text{mistake}) \Rightarrow \begin{aligned} R_1 &= \{x: p(x, c_1) > p(x, c_2)\} \\ R_2 &= \{x: p(x, c_2) > p(x, c_1)\} \end{aligned}$$

Proof for this fact.

(在永远相同的overlap region)

Let x^* be arbitrary one,

$$\begin{cases} \text{when } x \in R_2 \rightarrow \text{the only contribution to } P(\text{mistake}) \text{ is } p(x^*, c_1) \\ \text{when } x \in R_1 \rightarrow \text{the only } \text{---} \text{ is } p(x^*, c_2) \end{cases}$$

$$\text{when } p(x^*, c_1) > p(x^*, c_2)$$

$$\Rightarrow \text{error} = p(x^*, c_2) \text{ is smaller} \Rightarrow x^* \in R_1$$

$$\therefore R_1 = \{x: p(x, c_1) > p(x, c_2)\}$$

Little Tips in the end.

when we write the decision region / rule,

$$\begin{aligned} \text{we can also write } p(c_1|x) > p(c_2|x) &\rightarrow \text{predict } C_1 \\ \text{or } R_1 &= \{x: p(c_1|x) > p(c_2|x)\} \end{aligned}$$

$$\text{since } p(c_1|x) > p(c_2|x) \iff p(c_1, x) > p(c_2, x)$$

$$\iff \frac{p(c_1, x)}{p(x)} > \frac{p(c_2, x)}{p(x)}$$

Expected Loss

离散情况下

问题: 我们做每种决策带来的后果可能不同. (更偏向做某些决策)

Solution: We want a loss function to measure the loss incurred by taking any of the available decisions.

In discrete case, we have a loss matrix.

$$\begin{pmatrix} 0 & L_{12} & L_{13} & \dots \\ L_{21} & 0 & \dots & \dots \\ \vdots & \vdots & 0 & \dots \\ \vdots & \vdots & \vdots & 0 \end{pmatrix} \quad \text{e.g.}$$

For every x , L_{kj} is the loss that the true class is C_k , but we assign it to class C_j . (or make decisions that $x \in R_j$)

Then, our new objective goes to

$$\begin{aligned} E[L] &= \sum_{k,j} L_{kj} P(x \in R_j, C_k) \\ &= \sum_k \sum_j L_{kj} \int_{R_j} p(x, C_k) dx = \sum_k \sum_j \int_{R_j} L_{kj} p(x, C_k) dx \end{aligned}$$

Tip in here

Misclassification Rate $\subset$ Expected Loss

In the last cancer example, $k \in \{1, 2\}$, $j \in \{1, 2\}$, Loss Matrix = $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\text{Then, } E[L] = \sum_k \sum_j L_{kj} P(x \in R_j, C_k)$$

$$= L_{11} P(x \in R_1, C_1) + L_{12} P(x \in R_2, C_1) + L_{21} P(x \in R_1, C_2) + L_{22} P(x \in R_2, C_2)$$

$$\begin{matrix} L_{11} = L_{22} \\ = 0 \end{matrix} \rightarrow = L_{12} P(x \in R_2, C_1) + L_{21} P(x \in R_1, C_2)$$

$$\begin{matrix} L_{12} \\ = L_{21} = 1 \end{matrix} \rightarrow = \int_{R_2} p(x, C_1) dx + \int_{R_1} p(x, C_2) dx = \boxed{P(\text{mistake})}$$

In this way, (recall: $E[L] = \sum_k \int_{R_j} p(x, c_k) L_{kj} dx$)

we want to choose $\{R_j\}_j$ that $\min_{\{R_j\}} E[L]$

具体做法:

$$\begin{aligned} E[L] &= \sum_k \int_{R_j} L_{kj} p(x, c_k) dx \\ &= \sum_j \int_{R_j} \sum_k L_{kj} p(x, c_k) dx \end{aligned}$$

Let $g_j(x) = \sum_k L_{kj} p(x, c_k)$ then, $E[L] = \sum_j \int_{R_j} g_j(x) dx$

$$\therefore E[L] = \int_{R_1} g_1(x) dx + \int_{R_2} g_2(x) dx + \dots$$

Then, it's obvious that $\min E[L]$ by R_j is

$$R_j = \{x: g_j(x) < g_i(x) \quad \forall i \neq j\}$$

[eg. 在 $\int_{R_1} g_1(x) dx$ 中, 如果 R_1 里面的 x 有 $\forall i \neq j, g_i(x) > g_j(x)$ 的修改,
则 $\int_{R_1} g_1(x) dx < \int_{R'_1} g_1(x) dx \quad \forall R'_1 \neq R_1$]

R'_1 可能有些 x 是 $g_2(x)$ 小,
但仍然对 $g_1(x)$ 较大

Simplifying Further:

$$R_j = \{x: g_j(x) < g_i(x) \quad \forall i \neq j\}$$

$$R_j = \{x: \sum_k L_{kj} p(x, c_k) < \sum_k L_{ki} p(x, c_k) \quad \forall i \neq j\}$$

$$\begin{aligned} p(x, c_k) \\ = p(c_k | x) p(x) \end{aligned}$$

$$R_j = \{x: \sum_k L_{kj} p(c_k | x) < \sum_k L_{ki} p(c_k | x) \quad \forall i \neq j\}$$

done!

Reject Option:

问题: 当两者都没有很有力, 差距很小但是一方有优势. e.g. $0.45 < 0.47$

Solution: Reject 也可以列入到分类方式中 e.g. $j \in \{1, 2, \text{reject}\}$

$$\left[\begin{array}{l} \text{e.g. } \text{Loss}(C_1) = g_1(x) = 8.9 \\ \text{Loss}(C_2) = g_2(x) = 9.0 \end{array} \right. \quad \text{Loss}(\text{reject}) = 2$$

we choose reject instead of making a classification decision.

连续情况下

问题: $t \in \mathbb{R}$ is continuous. 我们有 $p(x, t)$, 我们最后想要是一个值 (like $y(x_{\text{new}}) \approx t$), 现在没有 reject region 了, 怎么做决策?

Solution: 依旧 $\min E[L]$. (same as Bayes Risk Estimator)

$$\min_{y(x)} E_t[L] = \iint L(y(x), t) p(x, t) dx dt$$

小tip - classic when $L(y(x), t) = (y(x) - t)^2$ squared error loss.
 $y(x) = E[t|x]$

Proof:

$$\min_{y(x)} E_{x,t}[L] = \int \left[\int L(y(x), t) p(x, t) dt \right] dx$$

$$\therefore \forall x, \text{ we minimize } y(x) = \underset{y(x)}{\operatorname{argmin}} \int L(y(x), t) p(x, t) dt$$

$$= \underset{y(x)}{\operatorname{argmin}} \int L(y(x), t) p_t(x) dt \quad \leftarrow \text{divide } p(x)$$

$$\stackrel{\text{set } y=0}{=} \underset{y}{\operatorname{argmin}} \int (y - t)^2 p_t(x) dt$$

$$\Rightarrow y = \int t p_t(x) dt = E[t|x] \quad \square$$

Graphical Models

Defn. Joint Distributions

- 对于一个系统来说, 我们一般只描绘它的 joint distribution,

$$p(x_1, x_2, \dots, x_N)$$

- 然而如果 $x_i \in \{0, 1\}$, 我们需要 $2^N - 1$ 个参数才能描述

$$p(x_1, x_2, \dots, x_N)$$

tip: 这里是 $2^N - 1$ 因为一旦我们知道 $2^N - 1$ 个, 最后一位则是 $1 - \sum p(x_1, \dots, x_N)$

- 我们也可以使用 conditional distribution 表达

$$\prod_{j=1}^N p(x_j | x_1, x_2, \dots, x_{j-1}) = p(x_1, \dots, x_N)$$

e.g. $p(x_1, \dots, x_5) = p(x_5 | x_1, x_2, x_3, x_4) p(x_4 | x_1, x_2, x_3) p(x_3 | x_1, x_2) p(x_2 | x_1) p(x_1)$

Defn. Conditional Independence

- Assume there are N r.v.s $x_1, x_2, \dots, x_N$

For set $A \subset \{1, 2, 3, \dots, N\}$, $B \subset \{1, 2, 3, \dots, N\}$, $C \subset \{1, 2, 3, \dots, N\}$

A, B, C are disjoint and $\pi_A = \{x_i : i \in A\}$ $\pi_B = \{x_i : i \in B\}$...

we say ① $\pi_A \perp \pi_B | \pi_C$

iff ② ~~can~~ random variables π_A, π_B are conditionally independent given π_C

iff ③ $p(\pi_A, \pi_B | \pi_C) = p(\pi_A | \pi_C) p(\pi_B | \pi_C)$

$$p(\pi_A | \pi_B, \pi_C) = p(\pi_A | \pi_C)$$

$$p(\pi_B | \pi_A, \pi_C) = p(\pi_B | \pi_C)$$

Defn. Directed Acyclic Graphical Models (Bayes' Nets)
[DAG]

用处: 在一定 assumption 下, 缩减 joint distribution 的写法.
(加入一些 conditional independence)

• DAG $\longleftrightarrow$ Factorization of joint distribution

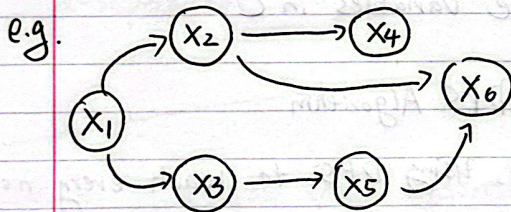
其中 nodes (of DAG) $\longleftrightarrow$ R.V.s

edges (of ...) $\longleftrightarrow$ dependence relationship

With a DAG, we can rewrite the joint distribution

$$p(x_1, \dots, x_N) = \prod_{j=1}^N p(x_j | x_1, \dots, x_{j-1}) = \prod_{i=1}^N p(x_i | \text{parents}(x_i))$$

tip: $\text{parents}(x_i)$ 代表 $\forall x_k$ s.t. $x_k \rightarrow x_i$



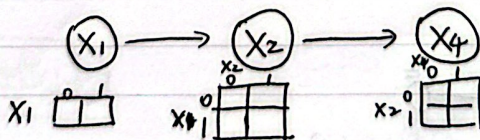
then,

$$p(x_1, x_2, \dots, x_6) = \prod_{i=1}^6 p(x_i | \text{parents}(x_i))$$

$$= p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2) p(x_5 | x_3) p(x_6 | x_2, x_5)$$

Defn. Conditional Probability Tables (CPT)

e.g.
Binary
Cases.



好处: 极大程度地减轻了 computational burden.

$\rightarrow$ with K_i parents

e.g. 2×2 CPT for x_4 只需 2 个参数
 $p(x_4 | x_2)$
 $p(x_4=0 | x_2=0)$
 $p(x_4=0 | x_2=1)$

$$\therefore N \text{ 个 CPT 需要 } \prod_{i=1}^N 2^{K_i} \leq \prod_{i=1}^N 2^{\max K_i}$$

$$= N 2^{\max K_i}$$

$$<<< 2^N - 1$$

Defn. D-Separation

Directed Separation — 一种描述在特定规则下, DAG图中,
在 conditioned on X_C 时, (X_A, X_B) 有没有连接.

is a notion of connectedness in DAGs in which two (sets of) variables may or may not be connected conditioned on a third (sets of) variables

性质: D-separation in DAG

$\Leftrightarrow$

Conditional Independence

具体来讲: $X_A \perp X_B \mid X_C$ Conditional independence
iff

every variables in A are d-separated from every variable
in B conditioned on all the variables in C.

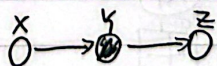
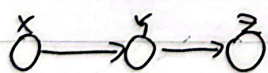
① 第一种检查 d-separation 的算法 — DFS Algorithm

- cycle through each node of A, using DFS to reach every node in B
- Examine all the path
- All paths are blocked

$\longleftrightarrow$ All paths / r.v.s are d-separated

$\longleftrightarrow X_A \perp X_B \mid X_C$

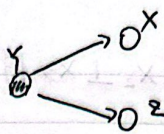
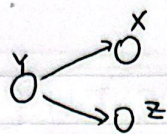
一些简单的 causal Chain 证明



$$p(z|x,y) = \frac{p(z,x,y)}{p(x,y)} = \frac{p(x)p(y|x)p(z|y)}{p(y|x)p(x)}$$

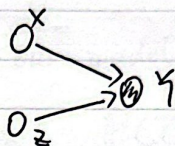
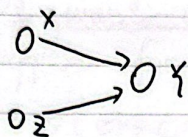
$$= p(z|y)$$

$\therefore$ 图像 X, Z are d-separated $\Leftrightarrow p(z|x,y) = p(z|y)$ 总计算
conditioned on X



$$p(x, z | y) = \frac{p(x, y, z)}{p(y)} = \frac{p(x|y) p(z|y) p(y)}{p(y)} = p(x|y) p(z|y)$$

图像 X, Z d-separated conditioned on $Y \iff p(x, z | y) = p(x|y) p(z|y)$ 对



$$p(z | x, y) = \frac{p(x, y, z)}{p(x, y)} = \frac{\cancel{p(x|z)} \cancel{p(y|z)} p(z)}{\cancel{p(x|z)} \cancel{p(y|z)} p(y)} = \frac{p(z) p(y | x, z)}{p(y | x)} \neq p(z | y)$$

图像 X, Z aren't d-separated $\iff p(z | x, y) \neq p(z | y)$

Defn. Bayes Ball Algorithm.

An algorithm for determining conditional independence in a DAG.

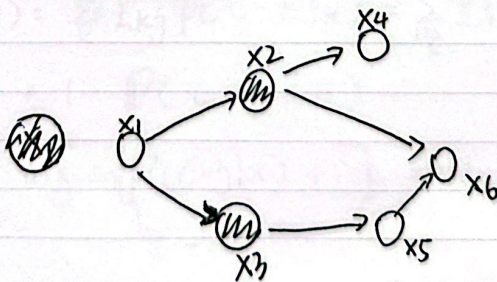
$$X_A \perp X_B | X_C$$

$\iff$

Every variable in A is d-separated from every variable in B conditioned on all variables in C .

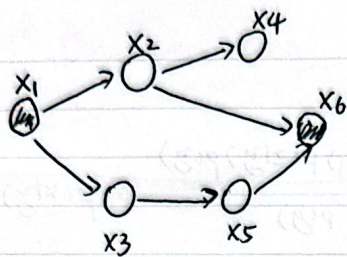
两个例子

e.g.



$$X_1 \perp X_6 | \{X_2, X_3\}$$

e.g.



$x_2 \perp x_3 \mid \{x_1, x_6\}$?

False!

$x_2 \rightarrow x_6 \rightarrow x_5 \rightarrow x_3$

TUT 2

Decisions and Bayes Ball

Decision Theory

Now we are in a discrete case, we have the joint distribution $p(x, c)$.

We want a decision.

e.g. Suppose for one input, our model gives

$$P(C=1|x) = 0.4 \quad P(C=0|x) = 0.6$$

defn. Conditional Risk

The average loss given a input x .

$$R(j|x) = E[L(C, j) | X=x] = \sum_k L_{kj} P(C=k | X=x) \quad \leftarrow \text{离散}$$

$$= \int L(C, j) f(C|x) dC \quad \leftarrow \text{连续}$$

注意: 跟 Bayes Estimator 很像.

$$r_{\pi}(\hat{\theta}) = \int R(\hat{\theta}, \theta) \pi(\theta) d\theta$$

$$\hat{\theta} = \operatorname{argmin}_{\theta} r_{\pi}(\hat{\theta}) = \operatorname{argmin}_{\theta} \int L(\hat{\theta}, \theta) \pi(\theta|x) d\theta$$

∴ 我们的 decision

→ Given x , 选 $j^*(x)$ s.t. $j^*(x) \in \operatorname{argmin}_j R(j|x)$

在特殊情况下,

$$\text{zero-one loss} \longleftrightarrow L_{kj} = \mathbb{1}\{k \neq j\}$$

$$R(j|x) = \sum_k L_{kj} P(C=k|x) = \sum_k \mathbb{1}\{k \neq j\} P(C=k|x) = \sum_{k \neq j} P(C=k|x)$$

$$= 1 - P(C=j|x)$$

$$\text{then } \operatorname{argmin}_j [P(C=j|x) + 1] = \operatorname{argmax}_j P(C=j|x)$$

∴ zero-one loss → 后验概率最大

继续例子

$$P(C=1|x) = 0.4 \quad P(C=0|x) = 0.6$$

	choose 0	choose 1
true 0	0	1
true 1	4	0

$$① R(0|x) = \sum_{k=0,1} L_{k0} P(C=k|x) = L_{10} P(C=1|x) = 4 \times 0.4 = 1.6$$

$$R(1|x) = \sum_{k=0,1} L_{k1} P(C=k|x) = L_{01} P(C=0|x) = 1 \times 0.6 = 0.6$$

$$② R(1|x) < R(0|x)$$

⇒ we will predict 1 although $P(C=1|x) = 0.4$ is smaller

③ 什么时候两者相等?

general: $q = P(C=1|x)$

$\begin{matrix} \swarrow L_{01} \\ \text{CFP: False-Positive Cost (choose 1, but it's 0)} \\ \text{CFN: False-Negative Cost (choose 0, but it's 1)} \\ \uparrow L_{10} \end{matrix}$

$$\therefore R(0|x) = L_{10} P(C=1|x) = \text{CFN} \cdot q$$

$$R(1|x) = L_{01} P(C=0|x) = \text{CFP} (1-q)$$

$$\begin{cases} R(0|x) > R(1|x) & \Rightarrow 1 \\ R(0|x) < R(1|x) & \Rightarrow 0 \\ \dots = \dots & \Rightarrow \text{reject / nothing.} \end{cases}$$

$$\therefore R(0|x) = R(1|x) \Rightarrow \text{CFN} \cdot q = \text{CFP} - \text{CFP} \cdot q$$

$$\Rightarrow q = \frac{\text{CFP}}{\text{CFN} + \text{CFP}}$$

when $q > \frac{\text{CFP}}{\text{CFN} + \text{CFP}} \Rightarrow \text{choose 1.}$

关于这个例子: $q = \frac{L_{01}}{L_{01} + L_{10}} = \frac{1}{1+4} = 0.2$ and $P(C=1|x) = 0.4 > 0.2$

继续例子

如果我们还有 reject option 的话呢?

Suppose zero-one loss and a fixed reject cost λ .

$$R(0|x) = L_{01} P(C=1|x) = q$$

$$R(1|x) = 1-q$$

$$R(\text{reject}|x) = \lambda$$

we always choose $j = \operatorname{argmin} R(j|x)$

$\therefore$ we will choose reject when $\lambda \leq q$ and $\lambda \leq 1-q$

$$\text{i.e. } \lambda \leq \min(q, 1-q)$$

$$\text{i.e. } q \leq 1-\lambda \quad 1-q \leq 1-\lambda$$

$$\text{i.e. } \max(q, 1-q) \leq 1-\lambda$$

$$\text{i.e. } \max_c P(C=c|x) \leq 1-\lambda$$

连续情况

$$\hat{j} = \operatorname{argmin}_j R(j|x) = \operatorname{argmin}_j \int L(t, j) f(t|x) dt$$

Thm. $L(t, j) = \text{squared loss} \iff \hat{j} = E[t|x]$

Pf: $\hat{j} = \operatorname{argmin}_j \int (t-j)^2 f(t|x) dt$

$$\frac{d}{dj} \int (t-j)^2 f(t|x) dt = \int t f(t|x) dt - j \int f(t|x) dt = 0$$

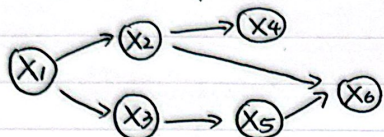
$$\Rightarrow \hat{j} = \int t f(t|x) dt = E[t|x]$$

End for decision theory.

Graphical Models

↓ 312

The graph specifies the model



$$p(x) = \prod_i p(x_i | \text{parent}(x_i))$$

eg. In this case

$$p(x_1, \dots, x_6) = p(x_1) p(x_2 | x_1) p(x_3 | x_1) p(x_4 | x_2) p(x_5 | x_3) p(x_6 | x_2, x_5)$$

Independence, d-separation and Bayes Ball

defn.

Independence $\longleftrightarrow$ probability distribution 的一种性质

D-separation $\longleftrightarrow$ 一种图论准则 (性质), 和 independence 对应

Bayes Ball $\longleftrightarrow$ 一种在图中判断 d-separation 的算法

defn. Conditional Independence

$$A \perp\!\!\!\perp B \mid Z \iff p(a, b \mid z) = p(a \mid z) p(b \mid z)$$

defn. D-separation

对于一套 path rule

Pattern	Path	Blocked when
Chain	$A \rightarrow M \rightarrow B$	M is observed
Fork	$A \leftarrow M \rightarrow B$	M is observed
Collider	$A \rightarrow M \leftarrow B$	M or any of its descendant is <u>not</u> observed

A path is active when there is no middle node

blocks it.

D-separation $\longleftrightarrow$ All paths are blocked.
/ no active path.

defn. Bayes Ball



对于V: 从P来 $\longleftrightarrow$ arrival from parent
从K来 $\longleftrightarrow$ arrival from child

The four Bayes Ball movement rules

Current node	Ball arrives from	Send the ball to
Unobserved	Parent	All children
Unobserved	Child	All parents and children
Observed	Parent	All parents
Observed	Child	Stop

算法过程

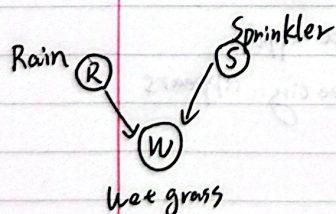
Objective: $A \perp\!\!\!\perp B \mid Z$?

1. Shade the evidence Z
2. Start at A, as arrival from child. the variables we observe / condition on
3. Apply the movement rules
4. Record. Skip the state already processed.

到达了 ——— d-connected / an active connection exists
没到达 ——— d-separated / all paths are blocked.

两个tip ——— 对于理解 d-separation 和 bayes ball 算法

- ① 为什么, observed node 会连接 parents (在 Bayes Ball 中)
/ 为什么 ——— 会在 collider 中打开 path (在 d-separation 定义中)
/ 为什么 parents 会 dependence 在 conditioned on 同一个 child node.

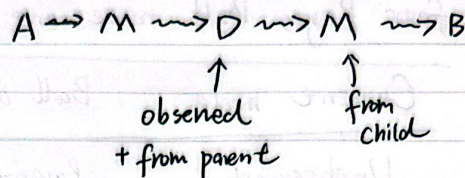
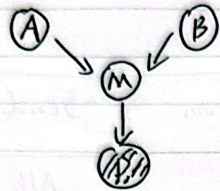


数学理解: $p(R|W,S) = p(R,W,S) / p(W,S)$

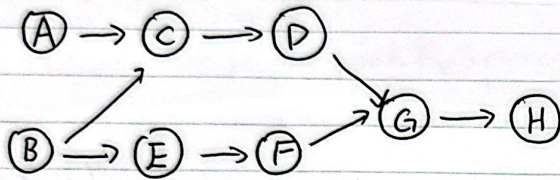
$$\begin{aligned}
 &= \frac{p(R)p(S)p(W|S,R)}{p(S)p(W|S)} \\
 &= \frac{p(R)p(W|S,R)}{p(W|S)} \neq p(R|W)
 \end{aligned}$$

感性: 知道 W 后,
已知下雨会让洒水 less likely
(explaining away)

- ② 为什么 d-separation 中, observed descender 也会打开 collider
(用 Bayes Ball 算法解释)



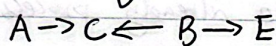
应用 Bayes Ball 的例子



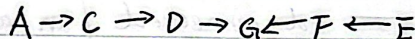
- ① $A \perp\!\!\!\perp E \mid Z = \emptyset$?

No.

Path 1

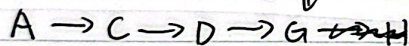


Path 2



- ② $A \perp\!\!\!\perp E \mid H$?

Yes.



现实建模小例子

$C=1$ spam

$C=0$ not spam

$X_1=1$ free app

$X_2=1$ meeting appears

Naive Bayes: $X_1 \perp\!\!\!\perp X_2 \mid C$

$$\Rightarrow p(X_1, X_2 \mid C) = p(X_1 \mid C) p(X_2 \mid C)$$

$$\therefore p(X_1, X_2, C) = p(X_1, X_2 \mid C) p(C) = p(X_1 \mid C) p(X_2 \mid C) p(C)$$

